Freedom ICT Care

(Dedicated For You)

Time: 30 Min Marks-20

1) $\boldsymbol{x+y+z=12,\ x^2+y^2+z^2=50\ and\ x+\frac{1}{x}=2}$

a) $Simplify: (4x+3y)^2+(4x-3y)^2+2(4x+3y)(4x-3y).$ 2

b) $Prove\ that\ x^2+\frac{1}{x^2}=x^4+\frac{1}{x^4}.$ 4

c) $Find\ the\ value\ of\ (x-y)^2+(y-z)^2+(z-x)^2.$ 4

2) $\boldsymbol{a^4+a^2b^2+b^4=8,\ a^2+ab+b^2=4\ and\ 2x+\frac{2}{x}=3}$

a) $Express\ a^2+10a+24\ as\ the\ difference\ of\ two\ squares.$ 2

b) $Find\ the\ value\ of\ x^2+\frac{1}{x^2}.$ 4

c) $Prove\ that\ 2(a^2+b^2)=6.$ 4

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